

TQM Tools and Techniques-II

(i) Concept of Quality Circle

There have been different interpretations of the concept of quality circles in various organizations. However, the most commonly accepted definitions in keeping with the essence of the philosophy as it originated in Japan are:

“Quality Circle is a small group of employees in the same work-area or doing a similar type of work who voluntarily meet regularly for about an hour every week to identify, analyze and resolve work-related problems, leading to improvement in their total performance, and enrichment of their work life” (Udupa 1986).

“Quality circles are a formal, institutionalized mechanism for productive and participative problem-solving interaction among employees” (Lozano & Thompson 1980).

“Quality control circle is not just a little room adjacent to the factory floor, whose occupants make a nuisance of themselves to everyone else. It is a state of mind and a matter of leadership with everyone from the president to production trainee involved” (Rehder 1981).

“Quality circle is a small group to perform capital quality control activities within the same workshop. This small group carries on continuously as a part of companywide quality control activities self-development and mutual development and improvement within the workshop, utilizing quality control techniques with all member participating” (Dr.Ishikawa).

Features of Quality Circles

The main features of quality circle are:

(a) Quality circle is a small group of employees

Quality circle is a small group of employees of 8 to 10. A circle with less than 5 members would lose its vitality due to high rate of absenteeism. This may cause a circle to become inactive. On the other hand, more than 15 members in a circle could result in denial of opportunity for active

participation by everyone. 8 to 10 are recommended as the minimum and maximum strength of quality circles respectively. The reason for such numbers is that number of interactions among members would be manageable.

(b) Quality circle is organized in the same work area or doing similar type of work

A quality circle is a homogeneous group and not an inter-departmental or inter-disciplinary one. Members participating in circle activities must be on the same wave-length. Discussions taking place at the meetings should be intelligible to each one of the members. This is possible only if the composition of the circle includes employees working in the same work area or engaged in a similar type of work. Designations of members need not necessarily be equal but the work in which they all are engaged should be common.

(c) Quality circles are voluntary

Employees decide to join quality circles on their own willingness. No compulsion, or pressure can be brought on any employee to join or not to join quality circles. This is based on voluntarism principle.

(d) Quality circles meet regularly for about an hour every week

Normally; a quality circle meets for about an hour every week. It is therefore possible for the circle to meet at least three or four times a month. The regularity of such meetings is very significant and it must be adhered to. These meetings could be conducted during or after working hours. This decision is left to quality circle members themselves.

(e) Quality circles identify, analyses and resolves work-related problems

The employees who work continuously in a work area knows best what problems are hindering achievement of high quality, productivity and optimum performance as also how they can be remedied. The members of quality circles themselves can, therefore, identify problems and entertain requests from the management and other departments to look into certain problems that may be worrying them. The focus of quality circles is “work related problems” and not other extraneous issues such as grievances or demands.

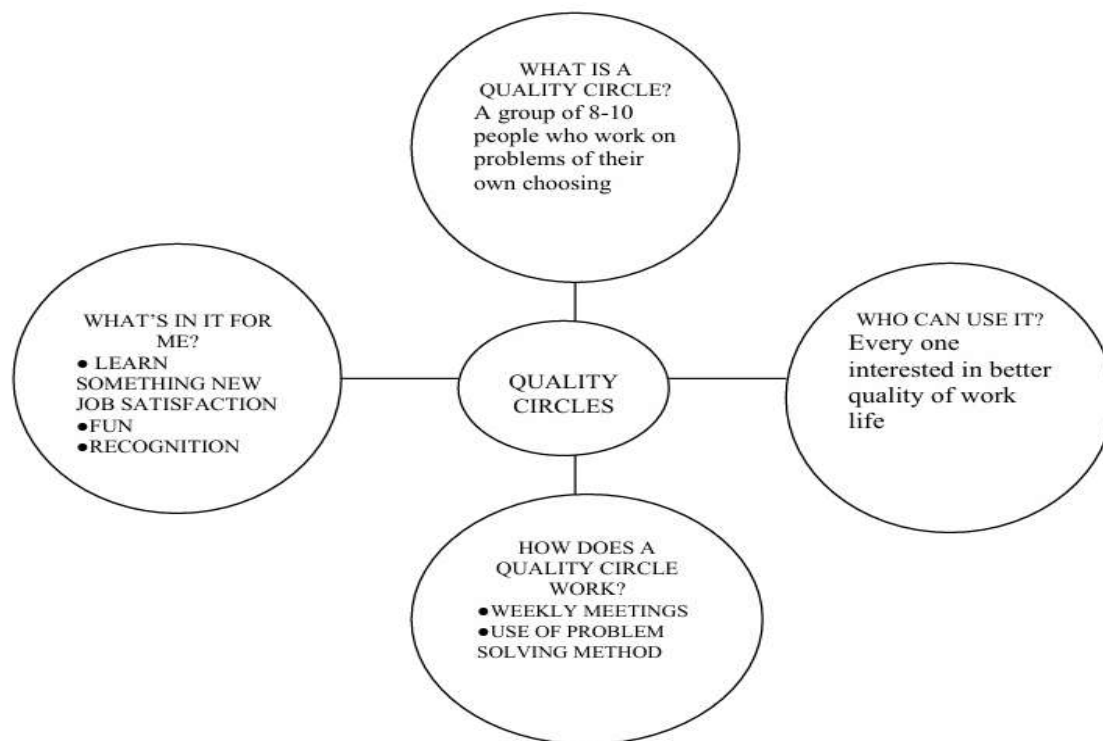
(f) Quality circle leads to total performance

As quality circles resolve work related problems relating to quality, productivity, cost reduction, safety etc. the total performance of the work area naturally improves. This results in both tangible and intangible gains to the whole organization.

(g) Quality circle enriches work life

The spin off benefits of quality circles of the organization includes enrichment of the work life of their employees apart from attitudinal changes, cohesive team culture, etc. Improved working environment, happier relations with co-employees, greater job satisfaction etc. are responsible for this enrichment of their work life.

Quality Circle in a nutshell is presented as-



(Source: Mathew George 1991)

(ii) Quality Function Development

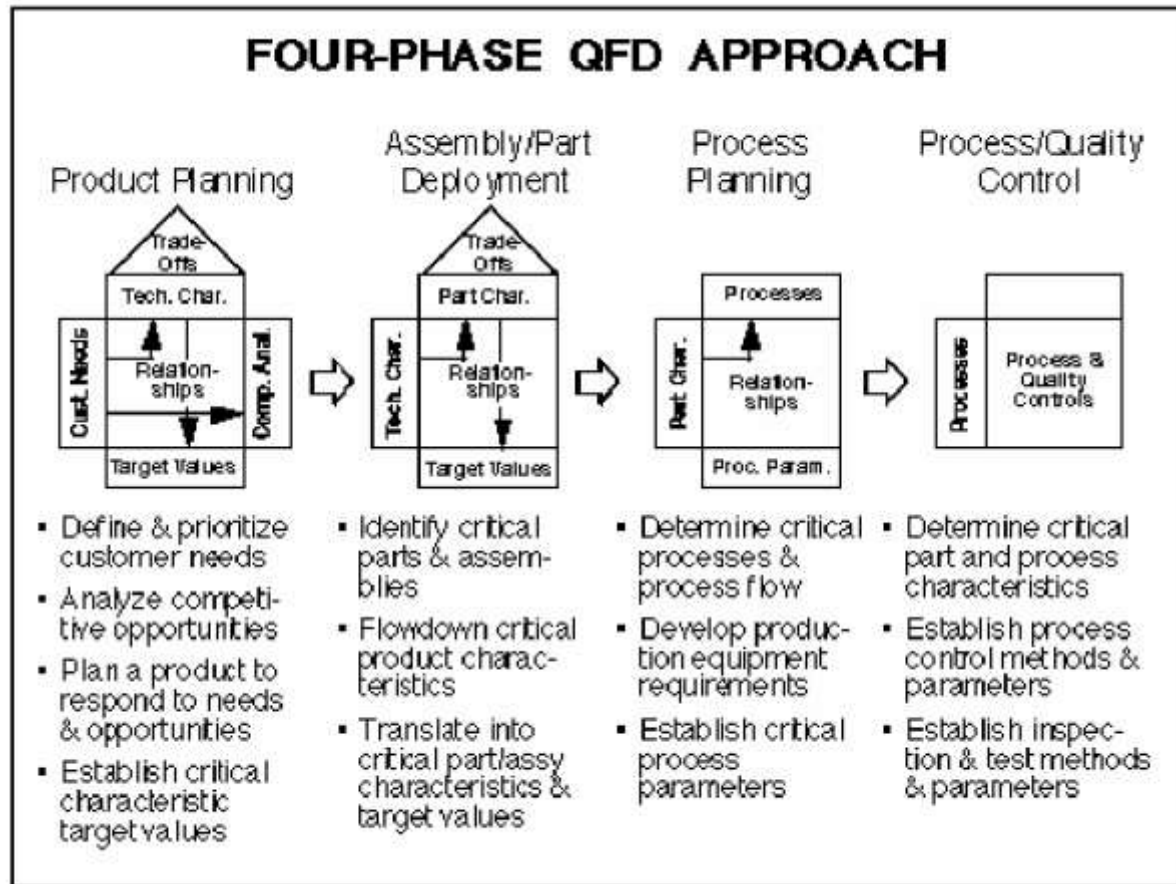
Quality must be designed into the product, not inspected into it. Quality can be defined as meeting customer needs and providing superior value. This focus on satisfying the customer's needs places an emphasis on techniques such as Quality Function Deployment to help understand those needs and plan a product to provide superior value.

Quality Function Deployment (QFD) is a structured approach to defining customer needs or requirements and translating them into specific plans to produce products to meet those needs. The "voice of the customer" is the term to describe these stated and unstated customer needs or requirements. The voice of the customer is captured in a variety of ways: direct discussion or interviews, surveys, focus groups, customer specifications, observation, warranty data, field reports, etc. This understanding of the customer needs is then summarized in a product planning matrix or "house of quality". These matrices are used to translate higher level "what's" or needs into lower level "how's" - product requirements or technical characteristics to satisfy these needs.

While the Quality Function Deployment matrices are a good communication tool at each step in the process, the matrices are the means and not the end. The real value is in the process of communicating and decision-making with QFD. QFD is oriented toward involving a team of people representing the various functional departments that have involvement in product development: Marketing, Design Engineering, Quality Assurance, Manufacturing/ Manufacturing Engineering, Test Engineering, Finance, Product Support, etc.

QFD METHODOLOGY FLOW

The basic Quality Function Deployment methodology involves four basic phases that occur over the course of the product development process. During each phase one or more matrices are prepared to help plan and communicate critical product and process planning and design information. This QFD methodology flow is represented below.



Quality Function Deployment begins with product planning; continues with product design and process design; and finishes with process control, quality control, testing, equipment maintenance, and training. As a result, this process requires multiple functional disciplines to adequately address this range of activities. QFD is synergistic with multi-function product development teams. It can provide a structured process for these teams to begin communicating, making decisions and planning the product. It is a useful methodology, along with product development teams, to support a concurrent engineering or integrated product development approach. Quality Function Deployment, by its very structure and planning approach, requires that more time be spent upfront in the development process making sure that the team determines, understands and agrees with what needs to be done before plunging into design activities. As a result, less time will be spent downstream because of differences of opinion over design issues or redesign because the product was not on target. It leads to consensus decisions, greater commitment to the development effort, better coordination, and reduced time over the course of the development effort. QFD requires discipline. It is not necessarily easy to get started with.

(iii) PM Concept

Project management is concerned with the overall planning and co-ordination of a project from conception to completion aimed at meeting the stated requirements and ensuring completion on time, within cost and to required quality standards.

Project management is normally reserved for focused, non-repetitive, time-limited activities with some degree of risk and that are beyond the usual scope of operational activities for which the organization is responsible.

Steps in Project Management

The various steps in a project management are:

1. Project Definition and Scope
2. Technical Design
3. Financing
4. Contracting
5. Implementation
6. Performance Monitoring

Project Definition and Scope What is a Project?

"A project is a one-shot, time-limited, goal-directed, major undertaking, requiring the commitment of varied skills and resources".

A project is a temporary endeavor undertaken to create a unique product or service. A project is temporary in that there is a defined start (the decision to proceed) and a defined end (the achievement of the goals and objectives). Ongoing business or maintenance operations are not projects. Energy conservation projects and process improvement efforts that result in better business processes or more efficient operations can be defined as projects. Projects usually include constraints and risks regarding cost, schedule or performance outcome.

Four Basic Elements of Project Management

A successful Project Manager must simultaneously manage the four basic elements of a project: resources, time, cost, and scope. Each element must be managed effectively. All these elements are interrelated and must be managed together if the project, and the project manager, is to be a success.

(i) Managing Resources

A successful Project Manager must effectively manage the resources assigned to the project. This includes the labor hours of the project team. It also includes managing labor subcontracts and vendors. Managing the people resources means having the right people, with the right skills and the proper tools, in the right quantity at the right time.

However, managing project resources frequently involves more than people management. The project manager must also manage the equipment (cranes, trucks and other heavy equipment) used for the project and the material (pipe, insulation, computers, manuals) assigned to the project.

(ii) Managing Time and Schedule

Time management is a critical skill for any successful project manager. The most common cause of bloated project budgets is lack of schedule management. Fortunately, there is a lot of software on the market today to help you manage your project schedule or timeline.

Any project can be broken down into a number of tasks that have to be performed. To prepare the project schedule, the project manager has to figure out what the tasks are, how long they will take, what resources they require, and in what order they should be done.

(iii) Managing Costs

Often a Project Manager is evaluated on his or her ability to complete a project within budget. The costs include estimated cost, actual cost and variability. Contingency cost takes into account influence of weather, suppliers and design allowances.

How the 80/20 Rule can help a project manager?

The 80/20 Rule means that in anything a few (20 percent) are vital and many (80 percent) are trivial. Successful Project Managers know that 20 percent of the work (the first 10 percent and the last 10 percent) consumes 80 percent of your time and resources.

Project Management Life Cycle

The process flow of Project management processes is shown in Figure. The various elements of project management life cycle are

- a) Need identification
- b) Initiation
- c) Planning
- d) Executing
- e) Controlling
- f) Closing out

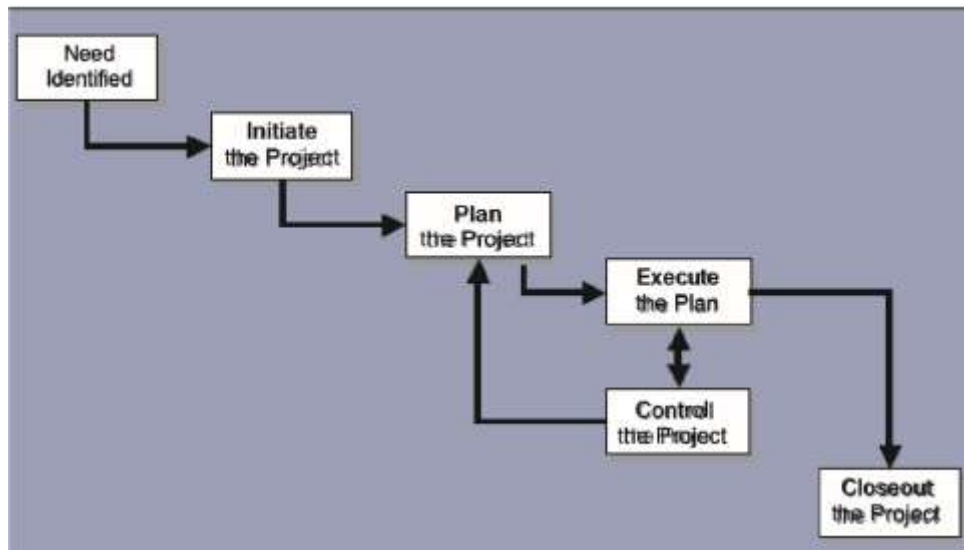


Figure: Process Flow of a Project Management Process

a) Need Identification

The first step in the project development cycle is to identify components of the project. Projects may be identified both internally and externally:

- Internal identification takes place when the energy manager identifies a package of energy saving opportunities during the day-to-day energy management activities, or from facility audits.
- External identification of energy savings can occur through systematic energy audits undertaken by a reputable energy auditor or energy service company.

In screening projects, the following criteria should be used to rank-order project opportunities.

- Cost-effectiveness of energy savings of complete package of measures (Internal rate of return, net present value, cash flow, average payback)
- Sustainability of the savings over the life of the equipment.
- Ease of quantifying, monitoring, and verifying electricity and fuel savings.
- Availability of technology, and ease of adaptability of the technology to Indian conditions.
- Other environmental and social cost benefits (such as reduction in local pollutants, e.g. SO_x)

b) Initiation

Initiating is the basic processes that should be performed to get the project started. This starting point is critical because those who will deliver the project, those who will use the project, and those who will have a stake in the project need to reach an agreement on its initiation. Involving all stakeholders in the project phases generally improves the probability of satisfying customer requirements by shared ownership of the project by the stakeholders. The success of the project team depends upon starting with complete and accurate information, management support, and the authorization necessary to manage the project.

c) Planning

The planning phase is considered the most important phase in project management. Project planning defines project activities that will be performed; the products that will be produced, and describes how these activities will be accomplished and managed. Project planning defines each major task, estimates the time, resources and cost required, and provides a framework for management review and control. Planning involves identifying and documenting scope, tasks, schedules, cost, risk, quality, and staffing needs. The result of the project planning, the project

plan, will be an approved, comprehensive document that allows a project team to begin and complete the work necessary to achieve the project goals and objectives. The project plan will address how the project team will manage the project elements. It will provide a high level of confidence in the organization's ability to meet the scope, timing, cost, and quality requirements by addressing all aspects of the project.

d) Executing

Once a project moves into the execution phase, the project team and all necessary resources to carry out the project should be in place and ready to perform project activities. The project plan is completed and base lined by this time as well. The project team and the project manager's focus now shift from planning the project efforts to participating, observing, and analyzing the work being done. The execution phase is when the work activities of the project plan are executed, resulting in the completion of the project deliverables and achievement of the project objective(s). This phase brings together all of the project management disciplines, resulting in a product or service that will meet the project deliverable requirements and the customers need. During this phase, elements completed in the planning phase are implemented, time is expended, and money is spent.

e) Controlling

Project Control function that involves comparing actual performance with planned performance and taking corrective action to get the desired outcome when there are significant differences. By monitoring and measuring progress regularly, identifying variances from plan, and taking corrective action if required, project control ensures that project objectives are met.

f) Closing out

Project closeout is performed after all defined project objectives have been met and the customer has formally accepted the project's deliverables and end product or, in some instances, when a project has been cancelled or terminated early. Although, project closeout is a routine process, it is an important one. By properly completing the project closeout, organizations can benefit from lessons learned and information compiled. The project closeout phase is comprised of contract closeout and administrative closure.

(iv) BPR- BUSINESS PROCESS REENGINEERING

What is BPR

Generally, the topic of BPR involves discovering how business processes currently operate, how to redesign these processes to eliminate the wasted or redundant effort and improve efficiency, and how to implement the process changes in order to gain competitiveness. The aim of BPR, according to Sherwood-Smith (1994), is **“seeking to devise new ways of organizing tasks, organizing people and redesigning IT systems so that the processes support the organization to realize its goals”**.

It is argued by some researchers (for example, van Meel et al., 1994; MacIntosh and Francis, 1997; Peltu et al., 1996) that there is no commonly agreed definition of BPR. Peltu et al. consider that this lack of an accepted definition of BPR makes it difficult to assess the overall success or failure of its concept.

The following is their definition of BPR: ***[Reengineering is] the fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in critical, contemporary measures of performance, such as cost, quality, service and speed.***

Another BPR father, Davenport (1993), describes „business process redesign“ as: ... the analysis and design of workflows and processes within and between organizations. Business activities should be viewed as more than a collection of individual or even functional tasks; they should be broken down into processes that can be designed for maximum effectiveness, in both manufacturing and service environment.

These definitions suggest that we should concentrate on *processes* rather than functions (or structures) as the focus of the (re-)design and management of business activity. The definitions of the term „process“ by different researchers are also slightly different. For example, Hammer and Champy (1993) define a process as: ***a collection of activities that takes one or more kinds of input and creates an output that is of value to the customer.***

A process is a specific ordering of work activities across time and space, with a beginning, an end, and clearly identified inputs and outputs: a structure for action. And Warboys et al. (1999) define a process as:

A process is structured change, i.e. there is a pattern of events which an observer may recognise across different actual examples (or occurrences) of the process, or which may be made manifest, or implemented, in many different occurrences. In BPR, the process to be reengineered is the so-called business process.

However, we should note that BPR is concerned with **customer-orientation**. Thus, the outputs of business processes should not only achieve the company objectives, but also need to satisfy customer requirements. From these definitions we can conclude that business processes start and end with customers, and the value of business processes is dependent upon customers.

BPR, DSS and TQM

When discussing radical change in BPR, we find that BPR, **DSS (decision support systems)** and **TQM (total quality management)** have much common with each other. Firstly, they are all focusing on business processes. Arnott and O'Donnell (1994) characterize DSS as relevant to BPR as it was the first information system (IS) movement to explicitly focus on the fundamental redesign of business processes rather than on the efficient application of a new computer technology. Also, BPR and DSS have a common aim which is to improve business processes via radical change.

The most significant difference between BPR and DSS is the *scope* of analysis: BPR focuses on the whole organization whereas DSS focuses on one individual decision. BPR is also different from TQM in that BPR concentrates on major discrete changes to business processes, whereas TQM concentrates on minor continuous improvement to business processes. That is, the improvements in TQM are smaller than the ones in BPR. Butler (1994) elucidates the difference between BPR and TQM as:

TQM approach which favors steady incremental gain, may often take a number of years to complete. For firms in highly competitive industries, this lag time can allow competitors to forge ahead. In contrast, results from BPR can be realized within 12-18 months, but it is a far riskier undertaking, and should not be regarded as a quick fix solution.

Furthermore, whereas BPR is commonly viewed as a top-down solution from management, TQM involves staff from all levels for problem solving and suggests bottom-up improvement. Employees' resistance to change has been identified as one major barrier to the success of BPR. It was reported by MacIntosh and Francis (1997) that those companies that had introduced TQM prior to taking on board BPR, faced less resistance to change. As we believe that these two approaches are compatible, we propose in this thesis a concept of „participative BPR“ which combines both of them.

(i) **Taguchi's Approach to Quality**

Taguchi defines quality in a negative way as ‘loss imparted to society from the time the product is shipped’. According to Taguchi, a product does not cause a loss only when it is outside specification limit but whenever it deviates from its target value. Any quality improvement programme should have its main objective, the minimization of the variation of product performance about its target value. Taguchi's recommended technique is a straightforward, well integrated system for implementing statistical experimental designs. It encourages proper experimentation and closer association between statisticians and engineers and increases statistical awareness in industry.

Professor, Genichi Taguchi, director of the Japanese Academy of Quality and four times recipient of the Deming Prize, devised a quality improvement technique that uses experimental design methods for efficient characterization of a product or process, combined with a statistical analysis of its variability. This approach allows quality considerations to be included at the early stage of any new venture: in the design and prototype phase for a product; during routine maintenance; or during installation and commissioning of a manufacturing process.

Taguchi's main objective lies in his definition of quality (or non-quality):

‘Quality is the amount of damage incurred to society from the moment the product has left the factory.’ Taguchi defines quality in a negative way as ‘loss imparted to society from the time the product is shipped’. By, ‘society’ he means the customer as well as the employees of the company; because, if the customer suffers, the whole company will eventually suffer through rework, guarantee cost, loss of reputation, loss of market share, and eventually loss of jobs.

Therefore, serving society is also Taguchi's aim.

Taguchi uses his loss function approach to establish a value base for the development of products. The function recognizes the need for average performance to match customer requirements and the fact that variability in this performance should be as small as possible. According to Taguchi, a product does not cause a loss only when it is outside specification limit but whenever it deviates from its target value. Any quality improvement programme should have its main objective the minimization of the variation of product performance about its target value. Smaller the performance variations better the quality. Larger the deviation from the target; larger the society's (producer's and consumer's) loss. This loss can be approximately evaluated by Taguchi's loss function, which unites with the function specification through a quadratic relationship.

Conventional Loss function and Taguchi's Loss Function

The traditional quality metric suggests

- ☐ All products within specifications are equally good
- ☐ All products outside specifications equally bad.

Unfortunately, this definition has led to the mindset which becomes a barrier towards improvement in products.

Figure 1: Traditional Loss Metric

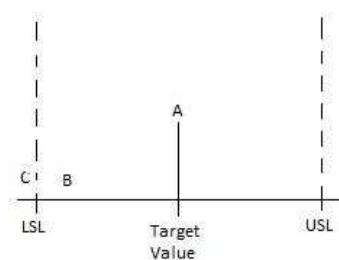


Figure 1 shows the target value, lower specification limit (LSL) and upper specification limit (USL). Both the products A and B are considered equally good although product A meets the specifications as per the target and product B is within the specification limit but far away from the target. Product C is considered to be non-acceptable or equally unacceptable as it is beyond the lower specification limit.

It was Taguchi who defined and quantified quality loss via his 'Loss Function' which is quite different from the traditional concept. He unites the financial loss with the functional specification through a quadratic relationship that comes from a Taylor Series Expansion (which is an approximation method).

Taguchi's loss function is as under:

$$L(Y) = M/D^2 (Y-t)^2$$

Where $L(Y)$ = the quality loss function t = target value/most desirable value of a product

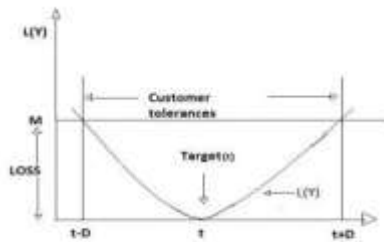
Y = performance of a product/the response found in test or analysis

M = producer's loss (in monetary terms)

D = customer's tolerance

The figure 2 shows the Taguchi's loss function

Figure 2: Graphical representation of Taguchi's Loss Function



Source: Ross (1988)

As per figure 2 the loss to the society increases as the performance characteristic deviates from the target value (t). In general, this loss is square of the deviation from the target. In the figure $t + D$ and $t - D$ are the functional limits beyond which 50% of system product needs customer maintenance.